

Meta-Learning Driven Few-Shot Knowledge Transfer with Dual-Stage Adaptive Data Replay for Cross-Domain Recommendation

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Abstract

Cross-domain recommendation (CDR) has emerged as a promising solution by effectively alleviating data sparsity by leveraging information from auxiliary domains. However, a major challenge in CDR is its dependence on predefined alignment rules (e.g., structural or distribution matching) to achieve cross-domain knowledge transfer, which impose fixed transfer patterns and lack the flexibility need for diverse cross-domain scenarios. Furthermore, most existing approaches still rely on coarse-grained representations. Knowledge transfer built upon imprecise representations can, even with improved alignment rules, instead lead to negative transfer in the target domain. To address these challenges and optimize recommendation efficacy, a new framework named meta-learning driven few-shot knowledge transfer with dual-stage adaptive data replay for cross-domain recommendation (MFACDR) is proposed. Specifically, a new meta-learning driven few-shot knowledge transfer method is proposed. This method leverages overlapping parts as anchors to guide the non-overlapping parts in autonomously exploring alignment rules through meta-learning, thus enabling few-shot knowledge transfer and flexible handling of different cross-domain scenarios. In addition, a dual-stage adaptive data replay mechanism is proposed, which enables fine-grained cross-domain adaptability and helps to mitigate negative transfer. Extensive experiments on three real-world datasets consistently demonstrate the superior effectiveness and robustness of the proposed MFACDR.

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CCS Concepts

• Information systems → Recommender systems.

Keywords

Cross-domain recommendation, Meta-learning, Few-shot knowledge transfer, Adaptive data replay

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1 Introduction

Recommender systems have become indispensable in e-commerce, social media, and online content platforms, as they help filter useful information from massive datasets and uncover users' interests and preferences. Traditional approaches have achieved success in mapping users and items into a latent space for personalized recommendation. However, these methods remain vulnerable to two drawbacks: data sparsity and cold start, which are detrimental to the accuracy and robustness of recommendation models.

To address these drawbacks, cross-domain recommendation (CDR) has emerged, drawing inspiration from transfer learning [24]. In contrast to single domain recommendation where user-item interactions are sparse, CDR leverages auxiliary information from a source domain to alleviate sparsity in the target domain. For example, a cross-domain model is developed by linking two rating matrices within the collaborative filtering (CF) framework [15]. EM-CDR is proposed based on a multi-layer perceptron (MLP) to learn entity characteristics across domains, thus alleviating data sparsity of the target domain by capturing nonlinear mapping relationships between entities [24].

Despite these advances, existing CDR methods still face two limitations. First, overlapping users typically constitute only a small



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fraction of the total user base. Solely depending on them for cross-domain knowledge transfer leads to limited information flow. Several studies proposed using non-overlapping users for cross-domain alignment. The commonly adopted approaches can be broadly divided into two categories: structural alignment, which aligns cross-domain user relations or subgraph structures to enable knowledge transfer [9, 41]; and distributional alignment, which aligns non-overlapping user groups at the distributional level [22, 25]. These methods have provided effective solutions for alleviating the sparsity of overlapping data.

Second, a key issue remains: such approaches often rely on predefined alignment rules, such as subgraph partitioning based on user interest distributions or neighbor similarities. While these rules may be effective in certain cases, they generally lack adaptability to diverse cross-domain settings. Second, even with improved alignment methods, most representation learning approaches remain coarse-grained. Such coarse modeling not only limits the precision of transfer but also amplifies the risk of negative transfer in the target domain. Moreover, models that introduce new data without proper replay mechanisms often suffer from catastrophic forgetting [11, 40], and existing feature refinement strategies still lack the ability to handle noisy or synthetic data effectively [20, 32].

These limitations raise two key research challenges: CH1: How to develop a flexible method that learn cross-domain alignment rules instead of relying on static, predefined patterns? CH2: How can fine-grained learning mechanisms be designed to dynamically balance new and historical knowledge while enhancing robustness against noise and synthetic data in cross-domain scenarios?

To address the abovementioned challenges, a new framework named meta-learning driven few-shot knowledge transfer with dual-stage adaptive data replay for cross-domain recommendation (MFACDR) is proposed in this study. Specifically, a meta-learning paradigm is introduced that decomposes the transfer process into an inner-loop and an outer-loop. In the inner-loop, overlapping parts are treated as anchors, and the model minimizes alignment errors between domains. This process allows the model to acquire meta-parameters that serve as a well-initialized starting point for transfer. In the outer-loop, these meta-parameters are adapted to non-overlapping parts in a few-shot manner. Beyond alignment, a dual-stage adaptive data replay mechanism is proposed to address the risks of negative transfer. By dynamically integrating new and historical data during training, this mechanism enhances fine-grained representation learning and improves the stability of knowledge transfer. Generally, the proposed framework has the following contributions:

- A new method of meta-learning driven few-shot knowledge transfer (MFT) is proposed, which exploits overlapping users to learn transferable meta-parameters through inner and outer-loop optimization, thereby extending alignment knowledge to non-overlapping users. Compared with predefined alignment rules, MFT uses overlapping users as learning samples to guide the model to learn appropriate cross-domain alignment rules.
- A new dual-stage adaptive data replay (DADR) mechanism is proposed to dynamically combine new data and historical data for collaborative training. This design explicitly mitigates catastrophic forgetting and enhances robustness against noisy or

synthetic data, thereby facilitating fine-grained representation learning and improving adaptability across domains.

- Extensive experiments on three real-world datasets are conducted in this study. Experimental results show that the proposed framework MFACDR achieves superior performance over state-of-the-art baselines.

2 Related Work

2.1 Cross-Domain Recommendation

Common cross-domain recommendation methods include domain adaption-based, mapping-based, auxiliary learning-based and so on [39, 46]. Domain adaption-based methods mainly achieve implicit feature transfer by minimizing the distribution difference between the source domain and the target domain in the feature space [34]. They usually use the relationship between social networks and rating preferences as a medium to connect the two domains [29, 38]. While these methods improve alignment at the distribution level, semantic discrepancies between domains may still hinder personalization in the target domain.

Mapping-based methods link two domains through the construction of an explicit mapping function between the source and target domains [1]. Early work explored cluster-level models that align latent factors of user groups across domains [6]. While later approaches such as EMCDR and DTCDDR [24, 44] directly learned user-level embeddings with neural networks to capture nonlinear mapping relations. Variants further introduced semi-supervised mechanisms to enhance the representation of domain-specific and shared features [13]. These methods highlight the importance of fine-grained user-level modeling but are still constrained by the quality of the mapping function.

As for auxiliary learning-based models, reinforcement learning has been applied to select and weight a user's multiple interests [33]. Dual learning have been integrated to jointly extract user preferences from multiple domains while maintaining relational consistency in their respective latent spaces [17]. In addition, Wasserstein path have been employed to model user preference trajectories in the embedding space [22]. Furthermore, knowledge graph-enhanced models have been proposed to introduce auxiliary semantic information, improving knowledge transfer [42].

Overall, while these methods have advanced CDR, they still face limitations. They rely on predefined alignment rules, such as clustering users based on interest distributions, or learning fixed mapping functions in a shared embedding space. While these rules may be effective in certain cases, they generally lack flexibility across diverse cross-domain settings.

2.2 Few-Shot Learning

Due to the high cost of data collection and manual annotation, researchers aim to learn transferable knowledge from only a few samples and generalize it to new tasks, a challenge often referred to as few-shot learning (FSL) [28]. FSL methods exhibit considerable utility in tackling the sparse issue in recommender systems [8] [16] [31]. At the same time, researchers are gradually integrating meta-learning into FSL. MAML [37] trains a single vector to initialize all weight vectors in the classification head, making

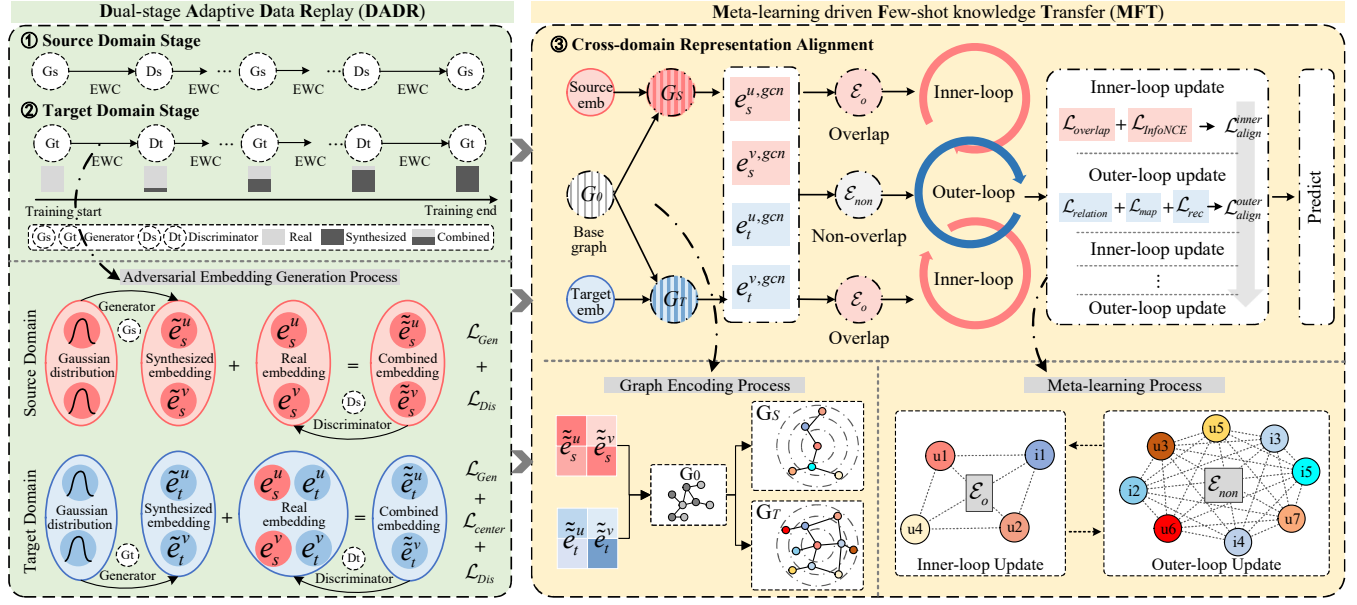


Figure 1: The architecture of the proposed MFACDR framework.

it suitable for small-shot learning. TML-KL [23] adopts a semi-supervised few-shot learning method with triple metric learning and Kullback-Leibler optimization.

Through a few gradient descent steps, a meta-learning-based model can be effectively fine-tuned into a robust model for any given few-shot task. This efficacy could be instrumental in resolving the issue of insufficient utilization of non-overlapping user data in cross-domain recommendation. Such an approach could change the predefined alignment rules by treating overlapping parts as few-shot samples, enabling the model to meta-learn adaptive alignment methods that can be generalized to non-overlapping parts.

2.3 Generative Adversarial Networks

The basic idea of adversarial generative networks (GAN) is to achieve data generation through the adversarial game between a generator and discriminator [7]. Based on their powerful data generation capability, many fields of research have incorporated GAN into their models [4] [12] [18].

Inspired by IRGAN [35], GAN began to be used in recommender systems. APR [10] enhanced the Bayesian personalized ranking (BPR) score with adversarial training, which improves model robustness and personalized ranking performance under implicit feedback. ECGAN-Rec [26] incorporated GAN into sequential recommendation with contrastive learning to model user-item interactions through adversarial training.

Within the field of CDR, a limited number of studies have similarly investigated the optimization of recommendation models by leveraging the data replay function of GAN. PPGenCDR [19] estimated the distribution of private data in the source domain based on GAN and solved the privacy protection problem in cross-domain recommendation. RecSys-DAN [34] adapted the GAN for data replay to alleviate data sparsity. This model maps user-item interaction data from two domains into a shared latent space, aiming

to make the discriminator indistinguishable from this data. Direct mapping may struggle to achieve perfect alignment when the distributions of the two domains differ significantly. Furthermore, using the same generative adversarial network for both domains may impair the model's ability to learn meaningful representations.

3 Methodology

3.1 Preliminary

In typical CDR scenarios, two domains are assumed to exist, i.e., a source domain $S = (U_S, V_S, \mathcal{E}_S)$ and a target domain $T = (U_T, V_T, \mathcal{E}_T)$. In these domains, $U, V, \mathcal{E}$ represent the user, item and edge sets, respectively. $\mathcal{E}_S \in \{0, 1\}^{|D_S| \times |V_S|}$ and $\mathcal{E}_T \in \{0, 1\}^{|U_T| \times |V_T|}$ are binary matrices, where an element $\mathcal{E}_{ij}$ being 1 indicates that an interaction exists between user $u_i \in U$ and item $v_j \in V$. The overlap between the user and item sets of the source and target domains is defined as U_o and V_o . Based on this, the user and item sets can be redefined: $U_S = \{U_S, U_o\}$ and $V_S = \{V_S, V_o\}$ for the source domain, with $U_T = \{U_T, U_o\}$ and $V_T = \{V_T, V_o\}$ for the target domain. Here, $U_S(V_S)$ and $U_T(V_T)$ represent the unique, non-overlapping parts of the two domains. Similarly, the edge set can also be divided into $\mathcal{E}_T = \{\mathcal{E}_T, \mathcal{E}_o\}$ and $\mathcal{E}_S = \{\mathcal{E}_S, \mathcal{E}_o\}$.

Given the observed interaction data from both domains, this study aims to transfer the rich interaction knowledge from the source domain to the target domain. The ultimate goal is to infer the preferences of users in the target domain and predict which items they will interact with. Formally, items $v_j \in V_T$ are to be recommended to a target user $u_i \in U_T$ by leveraging the interaction data from both domains, $\mathcal{E}_S$ and $\mathcal{E}_T$.

3.2 Overall Architecture

In this section, the technical details of proposed MFACDR framework are presented. As illustrated in Figure 1, the MFACDR consists

of two core modules. First, the proposed DADR mechanism performs different replay processes in the source and target domains. This dual-stage design enables fine-grained representation learning. Next, MFT method performs cross-domain representation alignment. By treating overlapping users as anchors, an inner–outer loop optimization process is employed: the inner-loop minimizes alignment errors on overlapping entities to extract transferable meta-parameters, which are then passed to the outer-loop to guide alignment for non-overlapping users, thereby enabling few-shot knowledge transfer across domains. Finally, the learned representations and their refined alignments are utilized to predict the interactions in target domain.

3.3 Representation Learning

In data-sparse cross-domain transfer scenarios, we incorporate synthetic data to increase sample coverage and smooth the distribution. The theoretical proof is in Appendix A. However, if synthetic data is introduced all at once, the model will be contaminated by immature synthetic samples, leading to catastrophic forgetting. Therefore, this study proposes the DADR mechanism. The adversarial embedding generation strategy is used to ensure the quality of the generated synthetic data, while the replay-aware optimization scheduling strategy is used to automatically adjust the proportion of synthetic samples.

3.3.1 Adversarial Embedding Generation. Adversarial embedding generation is designed in a dual-stage manner, where independent replay processes are established in the source and target domains.

• **Source domain stage.** The generator and the discriminator used in source domain stage are denoted by Gen_S and Dis_S , respectively. The real user and item embeddings are denoted by e_s^u and e_s^v , respectively. According to previous studies [7, 35], the initial input of Gen_S is the latent noise vector $z \in \mathbb{R}^d$ sampled from a standard multivariate Gaussian distribution, where d is the embedding dimension. Therefore, the process of generating the synthesized user embedding $\tilde{e}_s^u$ and item embedding $\tilde{e}_s^v$ can be defined as:

$$\tilde{e}_s^u = Gen_S^u(z), \tilde{e}_s^v = Gen_S^v(z) \quad (1)$$

where $Gen_S^u(\cdot)$ and $Gen_S^v(\cdot)$ represents the user and item embedding generator, respectively. Next, these two sets of embeddings are processed according to the replay-aware optimization scheduling strategy (in Section 3.3.2) to yield the combined embedding pairs $(\tilde{e}_s^u, \tilde{e}_s^v)$. The task of the generator is to adjust its parameters through the adversarial training so that the synthetic embedding pairs gradually approach the distribution of the real embedding pairs. So, the objective function of generator $\mathcal{L}_{Gen, S}$ is defined as:

$$\mathcal{L}_{Gen, S} = \mathbb{E}_z [\log(1 - Dis_S(\tilde{e}_s^u \| \tilde{e}_s^v))] \quad (2)$$

where $\mathbb{E}[\cdot]$ is the expectation operator, $\|$ is the concatenation operator, and $\log(\cdot)$ is the natural logarithm. The task of the discriminator is to distinguish whether the input pair is real or synthetic, thereby providing feedback to the generator. So, the objective function of discriminator $\mathcal{L}_{Dis, S}$ is defined as:

$$\mathcal{L}_{Dis, S} = \mathbb{E}_{(e_s^u, e_s^v) \sim S} [\log Dis_S(e_s^u \| e_s^v)] + \mathbb{E}_z [\log(1 - Dis_S(\tilde{e}_s^u \| \tilde{e}_s^v))] \quad (3)$$

where $(e_s^u, e_s^v) \sim S$ indicates that (e_s^u, e_s^v) is sampled from the source domain. The GAN loss $\mathcal{L}_{GAN}^S$ in the source domain consists of the discriminator and the generator loss, as shown:

$$\mathcal{L}_{GAN}^S = \lambda_{Dis, S} \cdot \mathcal{L}_{Dis, S} + \lambda_{Gen, S} \cdot \mathcal{L}_{Gen, S} \quad (4)$$

where $\lambda_{Dis, S}$ and $\lambda_{Gen, S}$ represent the weights of the discriminator and the generator loss, respectively.

• **Target domain stage.** The same as source domain stage, the synthetic user-item embedding pairs are defined as $(\tilde{e}_t^u, \tilde{e}_t^v)$, and the combined embedding pairs that are fed into the discriminator are defined as $\tilde{e}_t^u, \tilde{e}_t^v$. The difference from the source domain stage is that to stabilize adversarial training in the target domain and improve cross-domain embedding alignment, the discriminator Dis_T is trained on real target interactions and a subset of source-domain samples $\hat{S}_O = \{(u, v) \in S | u \in T \text{ or } v \in T\}$. This subset specifically includes source-domain user-item pairs that overlap with the target domain, thereby ensuring semantic consistency and fostering a more transferable decision boundary for the discriminator. Thus, the objective function of generator and discriminator $\mathcal{L}_{Gen, T}$ and $\mathcal{L}_{Dis, T}$ are defined as:

$$\mathcal{L}_{Gen, T} = \mathbb{E}_z [\log(1 - Dis_T(\tilde{e}_t^u \| \tilde{e}_t^v))] \quad (5)$$

$$\mathcal{L}_{Dis, T} = \mathbb{E}_{e_t^u, e_t^v \sim (T \cup \hat{S}_O)} [\log Dis_T(e_t^u \| e_t^v)] + \mathbb{E}_z [\log(1 - Dis_T(\tilde{e}_t^u \| \tilde{e}_t^v))] \quad (6)$$

where $(e_t^u, e_t^v) \sim (T \cup \hat{S}_O)$ indicates that (e_t^u, e_t^v) is sampled from the target domain and a subset of source domain's real data distribution.

Since the interaction data in the target domain are sparse and some source domain data are added, deploying Gen_T requires not only aligning the distribution of synthetic and real embeddings, but also enhancing stability through the center-aware alignment regularization term. Therefore, the target domain GAN loss $\mathcal{L}_{GAN}^T$ consists of the discriminator loss, the generator loss, and the center-aware alignment loss, as shown:

$$\mathcal{L}_{GAN}^T = \lambda_{Dis, T} \cdot \mathcal{L}_{Dis, T} + \lambda_{Gen, T} \cdot \mathcal{L}_{Gen, T} + \lambda_{center} \cdot \mathcal{L}_{center} \quad (7)$$

where $\lambda_{Dis, T}$, $\lambda_{Gen, T}$ and λ_{center} represent the weights of the discriminator, the generator and center-aware alignment regularization term loss, respectively. $\mathcal{L}_{center}$ is defined as:

$$\mathcal{L}_{center} = \frac{1}{2} \|\mu_{\tilde{e}_t} - \mu_{e_t}\|^2 - \frac{1}{2} d_c^2 \quad (8)$$

where μ_{e_t} and $\mu_{\tilde{e}_t}$ are the mean of real and synthetic embeddings, i.e., $\mu_{e_t} = \text{mean}(e_t^u \| e_t^v)$ and $\mu_{\tilde{e}_t} = \text{mean}(\tilde{e}_t^u \| \tilde{e}_t^v)$, and d_c is the tolerance threshold. This regularization term only penalizes excessive deviations between the average representations of the true and generated embeddings, thereby tolerating minor variations while ensuring semantic consistency. By introducing this soft constraint, the model can avoid overfitting to synthetic noise and be more robust to unstable signals during cross-domain adaptation.

3.3.2 Replay-aware Optimization Scheduling. The replay-aware optimization scheduling integrates two strategies: dynamic scheduling at the data level controls how synthetic samples are gradually mixed with real interactions, while elastic weight consolidation (EWC) regularization at the parameter level preserves critical knowledge against forgetting.

• **Data level.** Let the current epoch be t , and the maximum training cycle be T_{max} . For a batch of size N , the number of synthetic

samples is:

$$N_{syn} = \left\lfloor \frac{t}{T_{max}} \cdot N \right\rfloor \quad (9)$$

where $\lfloor \cdot \rfloor$ represents the floor operation. For each batch, we concatenate N_{syn} embedding pairs of $(\tilde{e}_x^u, \tilde{e}_x^v)$, with $N - N_{syn}$ real embedding pairs (e_x^u, e_x^v) where $x \in \{s, t\}$ indicates the source or target domain. The mixed embeddings $(\tilde{e}_x^u, \tilde{e}_x^v)$ are then fed into the generator and discriminator for adversarial training.

• **Parameter level.** While the data-level scheduling controls the quantity, the parameter-level scheduling controls the quality of parameter updates. Let $\theta \in \{e_s^u, e_t^u, e_s^v, e_t^v, W_{Gen}, W_{Dis}\}$ represents the set of all trainable parameters, including the user embeddings, item embeddings and the weights of the generator and discriminator. During the training, all parameters can be shown as a vector $\theta = [\theta_1, \theta_2, \dots, \theta_n]$, where θ_i represents the current value of the parameter. Correspondingly, $\theta^* = [\theta_1^*, \theta_2^*, \dots, \theta_n^*]$ represents a set of previous parameters and θ_i^* represents the historical optimal value of the i th parameter. According to [14], the importance of each parameter θ_i is related to the F_i of the Fisher information matrix, as shown:

$$F_i = \mathbb{E}_{x \sim \tau_{old}} \left[\left(\frac{\partial \log p(x|\theta)}{\partial \theta_i} \right)^2 \right] \quad (10)$$

where τ_{old} represents the empirical distribution of historical interactions. This distribution is obtained based on the interaction in the previous stage. Each interaction $x = (u, v)$ represents an interaction pair, and the $p(x|\theta)$ is actually a scoring function which corresponds to the interaction probability. Here we use the BPR loss. Then, the EWC loss is shown as:

$$\mathcal{L}_{EWC} = \frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_i^*)^2 \quad (11)$$

where λ adjusts the regularization strength. Note that the EWC loss in the source and target domains is denoted as $\mathcal{L}_{EWC}^S$ and $\mathcal{L}_{EWC}^T$, respectively. In summary, the final source domain loss $\mathcal{L}_{total}^S$ and target domain loss $\mathcal{L}_{total}^T$ can be denoted as follow:

$$\mathcal{L}_{total}^S = \mathcal{L}_{rec}^S + \lambda_{EWC} \cdot \mathcal{L}_{EWC}^S + \lambda_{GAN} \cdot \mathcal{L}_{GAN}^S \quad (12)$$

$$\mathcal{L}_{total}^T = \mathcal{L}_{rec}^T + \lambda_{EWC} \cdot \mathcal{L}_{EWC}^T + \lambda_{GAN} \cdot \mathcal{L}_{GAN}^T \quad (13)$$

where $\mathcal{L}_{rec}^S$ and $\mathcal{L}_{rec}^T$ represent the recommendation task loss. Here we use the BPR loss. λ_{EWC} and λ_{GAN} represent the weights of EWC loss and GAN loss, respectively.

3.4 Cross-domain Representation Alignment

To realize effective cross-domain representation alignment, MFT adopts a meta-learning paradigm that decomposes the transfer process into an inner-loop and an outer-loop. Initially, raw data is processed into graph structures to enable GCN-based extraction. First, build an adjacency matrix A_0 , including all users and projects in both domains. This matrix is a binary matrix, 1 if there is interactive record, and 0 if there is no. Then, the matrix is normalized to obtain the basic graph G_0 , i.e., $G_0 = D^{-\frac{1}{2}} A_0 D^{-\frac{1}{2}}$, where D is the diagonal degree matrix. This base graph G_0 is constructed as a shared structure onto which domain-specific features are projected.

Given the user and item embeddings $\tilde{e}_s^u, \tilde{e}_s^v, \tilde{e}_t^u, \tilde{e}_t^v$, the graph of the source and target domains can be obtained, as shown:

$$G_S = G_0 \cdot (\tilde{e}_s^u \|\tilde{e}_s^v), G_T = G_0 \cdot (\tilde{e}_t^u \|\tilde{e}_t^v) \quad (14)$$

After segmenting the data set processed by the graph structure, the user and item representations of the two domains can be obtained, as shown:

$$e_s^{u, gcn} = G_S[1, \dots, |U|], \quad e_s^{v, gcn} = G_S[|U| + 1, \dots] \quad (15)$$

$$e_t^{u, gcn} = G_T[1, \dots, |U|], \quad e_t^{v, gcn} = G_T[|U| + 1, \dots] \quad (16)$$

where $|U|$ represents the length of user set.

3.4.1 Inner-loop Update. In the inner-loop optimization, the focus is placed on aligning the embeddings of overlapping entities. In detail, a set $\mathcal{E}_o$ where each tuple involves an interaction between a user $\hat{u} \in U_o$ and an item $\hat{v} \in V_o$ is sampled. These interactions serve as the basis, allowing the model to calculate the necessary ingredients for updating. The inner-loop loss function consists of two components: the overlapping loss $\mathcal{L}_{overlap}$, and the InfoNCE loss $\mathcal{L}_{InfoNCE}$. The $\mathcal{L}_{overlap}$ encourages the consistency of overlapping users and items in two domains. The calculation process is shown:

$$\begin{aligned} \mathcal{L}_{overlap} = & \sum_{\hat{u} \in U_o} \|f_{map}^{inner}(e_s^{\hat{u}, gcn}) - e_t^{\hat{u}, gcn}\|_2^2 \\ & + \sum_{\hat{v} \in V_o} \|f_{map}^{inner}(e_s^{\hat{v}, gcn}) - e_t^{\hat{v}, gcn}\|_2^2 \end{aligned} \quad (17)$$

where f_{map}^{inner} represents the mapping function and MLP is used, and $\hat{u}$ represents the $\hat{u}$ th overlapped user and $\hat{v}$ represents the $\hat{v}$ th overlapped item. The InfoNCE loss promotes alignment by maximizing the consistency between positive pairs (i.e. overlapping entities) and comparing them with all other embeddings. The calculation process is shown:

$$\begin{aligned} \mathcal{L}_{InfoNCE} = & -\frac{1}{U_o} \sum_{\hat{u} \in U_o} \log \frac{\exp(\text{sim}(e_s^{\hat{u}, gcn}, e_t^{\hat{u}, gcn})/\tau)}{\sum_{\hat{u}' \in U_o} \exp(\text{sim}(e_s^{\hat{u}, gcn}, e_t^{\hat{u}', gcn})/\tau)} \\ & -\frac{1}{V_o} \sum_{\hat{v} \in V_o} \log \frac{\exp(\text{sim}(e_s^{\hat{v}, gcn}, e_t^{\hat{v}, gcn})/\tau)}{\sum_{\hat{v}' \in V_o} \exp(\text{sim}(e_s^{\hat{v}, gcn}, e_t^{\hat{v}', gcn})/\tau)} \end{aligned} \quad (18)$$

where $\hat{u}'$ and $\hat{v}'$ represent unrelated users and items, i.e., negative entity, $\text{sim}(\cdot)$ represents the cosine similarity, and τ is the temperature hyperparameter. The total alignment loss in the inner-loop $\mathcal{L}_{align}^{inner}$ is defined as:

$$\mathcal{L}_{align}^{inner} = \lambda_{overlap} \cdot \mathcal{L}_{overlap} + \lambda_{Info} \cdot \mathcal{L}_{InfoNCE} \quad (19)$$

where $\lambda_{overlap}$ and λ_{Info} represent the weights of two components, respectively.

Based on this loss function, the model's parameters, comprising the overlapped user embeddings $e_s^{\hat{u}, gcn}, e_t^{\hat{u}, gcn}$, overlapped item embeddings $e_s^{\hat{v}, gcn}, e_t^{\hat{v}, gcn}$, and the mapping function f_{map}^{inner} , are updated through the training. The gradient descent update for each of these parameters $p^{inner} \in \{e_s^{\hat{u}, gcn}, e_t^{\hat{u}, gcn}, e_s^{\hat{v}, gcn}, e_t^{\hat{v}, gcn}, f_{map}^{inner}\}$ is calculated by minimizing $\mathcal{L}_{align}^{inner}$, as shown:

$$\tilde{p}^{inner} = p^{inner} - \eta \nabla_{p^{inner}} \mathcal{L}_{align}^{inner}(p^{inner}; \mathcal{E}_o) \quad (20)$$

where $\tilde{p}^{inner}$ represents updated parameters after inner-loop, and η represents the inner-loop learning rate.

3.4.2 Outer-loop Update. The outer-loop optimizes the model by applying the knowledge learned in the inner-loop to non-overlapping data. Let $\mathcal{E}_{non} = \mathcal{E}_s \cup \mathcal{E}_t$ represent a the interactions involving only non-overlapping users and items. The outer-loop loss function consists of three components: cross-domain mapping loss, recommendation loss and relation-based alignment loss. The cross-domain mapping loss is specifically designed to train cross-domain mapping functions that project embeddings from the source into the target. The calculation process is shown:

$$\mathcal{L}_{map} = \|\mathcal{f}_{map}^{outer}(e_s^{u,gcN}) - e_t^{u,gcN}\|_2^2 + \|\mathcal{f}_{map}^{outer}(e_s^{v,gcN}) - e_t^{v,gcN}\|_2^2 \quad (21)$$

The recommendation loss is used to test the current recommendation accuracy. The calculation process is shown:

$$\mathcal{L}_{rec} = \mathcal{L}_{BPR}(e_t^{u,gcN}; e_t^{v,gcN}) \quad (22)$$

where $\mathcal{L}_{BPR}$ represents we use BPR as the test function.

The user-item interactions between the two domains are different, but they may share similar interaction patterns. Computing the similarity of these interaction patterns helps align the embeddings of source domain users and target domain items, maintaining consistency in their interaction patterns. Therefore, we adopted relation-based alignment loss in outer-loop, as show:

$$\mathcal{L}_{relation} = \mathcal{L}_{BCE}(\mathcal{R}(e_s^{u,gcN}, e_t^{v,gcN}); y) \quad (23)$$

where $\mathcal{L}_{BCE}(\hat{y}; y)$ is the binary cross-entropy loss, which measures the gap between the probability output $\hat{y}$ of the model and the true label y , y denotes whether there is or not an interaction, and $\mathcal{R}(\cdot, \cdot)$ is defined as show:

$$\mathbf{h}_0 = [\mathcal{f}_{map}^{outer}(e_s^{u,gcN})\|e_t^{v,gcN}\|(\mathcal{f}_{map}^{outer}(e_s^{u,gcN}) - e_t^{v,gcN})] \quad (24)$$

$$\mathcal{R}(e_s^{u,gcN}, e_t^{v,gcN}) = \mathbf{W}_3 \cdot \text{ReLU}(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \cdot \mathbf{h}_0 + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3 \quad (25)$$

where $\mathbf{h}_0$ is feature matrix, $\mathbf{W}_1$, $\mathbf{W}_2$ and $\mathbf{W}_3$ are weight matrices, $\mathbf{b}_1$, $\mathbf{b}_2$ and $\mathbf{b}_3$ are bias matrices, and $\text{ReLU}(\cdot)$ is the activation function. The total alignment loss in the outer-loop $\mathcal{L}_{align}^{outer}$ is defined as:

$$\mathcal{L}_{align}^{outer} = \lambda_{map} \cdot \mathcal{L}_{map} + \lambda_{rec} \cdot \mathcal{L}_{rec} + \lambda_{relation} \cdot \mathcal{L}_{relation} \quad (26)$$

where, λ_{map} , λ_{rec} , and $\lambda_{relation}$ represent the weights of three components, respectively.

The outer-loop uses the meta-parameters from the inner-loop and adjusts them to ensure consistent alignment for non-overlapping data, which means that the initial value of the outer-loop parameters $p^{outer} \in \{e_s^{u,gcN}, e_t^{u,gcN}, e_s^{v,gcN}, e_t^{v,gcN}, \mathcal{f}_{map}^{outer}\}$ is $\tilde{p}^{inner}$. The outer-loop then adjusts the parameters p^{outer} by minimizing the prediction loss on $\mathcal{E}_{non}$, as shown:

$$\begin{aligned} \min_{p^{outer}} \mathcal{L}_{align}^{outer}(\tilde{p}^{outer}; \mathcal{E}_{non}) = \\ \min_{p^{outer}} \mathcal{L}_{align}^{outer}(p^{outer} - \eta \nabla_{p^{outer}} \mathcal{L}_{align}^{inner}(p^{outer}; \mathcal{E}_o); \mathcal{E}_{non}) \end{aligned} \quad (27)$$

where $\tilde{p}^{outer}$ represents updated parameters after outer-loop. This equation establishes a two-level optimization problem. By repeatedly alternating updates between overlapping and all entity pairs, this few-shot learning method effectively extends limited alignment knowledge from overlapping entities to the entire cross-domain

structure, resulting in robust and generalizable feature transfer. The theoretical proof is in Appendix B.

4 Experiments and Analysis

In this section, extensive experiments on real-world datasets are performed to address the following research questions:

Q1: How does the proposed MFACDR perform compared to state-of-the-art cross-domain methods?

Q2: What are the impacts of the DADR mechanism and the MFT method on the performance of MFACDR?

Q3: How is the robustness of the proposed MFACDR?

Q4: Why does MFACDR achieve superior performance?

4.1 Experimental settings

4.1.1 Datasets and Evaluation Metrics. Consistent with prior studies in this research field [21][43], experimental datasets were drawn from the MovieLens¹ and Amazon² platforms. The comprehensive information of datasets is shown in Appendix C. To evaluate model performance, three tasks were constructed with varied overlap ratios between users and items. Specifically, Task 1 (item overlap) uses the MovieLens-1M dataset (ML-1M), Task 2 (user overlap) uses the Amazon Book-to-Movie dataset (Book-Movie), and Task 3 (both user and item overlap) uses the Amazon Electronics-to-Video dataset (Elect-Video). To ensure data quality and reduce noise, users and items with fewer than 10 recorded interactions were removed from each dataset prior to experimentation [21][20].

The standard evaluation metrics for Top-K recommendation are adopted in this study [2][27], including recall (Recall@K), mean reciprocal rank (MRR@K), normalized discounted cumulative gain (NDCG@K), and hit (Hit@K). The experimental results are all the average results after running 5 times.

4.1.2 Baselines. To verify the superiority of the proposed MFACDR framework, this study selected the following state-of-the-art baseline models, including: CMF [30], EMCDR [24], CoNet [11], DCD-CSR [45], DTCDD [44], SSCDR [13], DisenCDR [3], DML [17], BiT-GCF [21], GDCCDR [20], and Fed-CLR [36]. The detail of the baseline models is in Appendix D.

4.1.3 Implementation Details. To ensure a fair comparison, the datasets are randomly split into training, validation, and test sets with a ratio of 8:1:1. For the proposed MFACDR framework, the training process is structured into four stages: source domain learning, target domain learning, cross-domain alignment, and target domain updates. The number of epochs for each stage is set to 100, 100, 50, and 100, respectively. The proposed framework is trained with a batch size of 2048, using the Adam optimizer with a learning rate of 0.001. Early stopping with a patience of 10 is applied to prevent overfitting. All experiments were conducted using PyTorch 2.4.1 on a single NVIDIA GeForce RTX 4070 GPU with 32 GB of RAM. For performance evaluation, we use Top-10.

4.2 Performance Comparison (Q1)

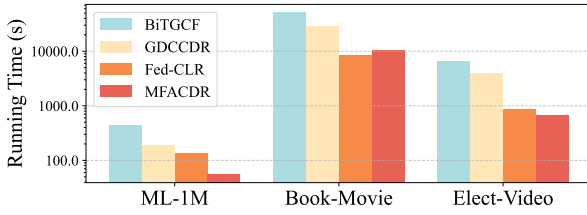
To address the research question Q1, this study selected 11 baseline models and conducted comparative experiments with the proposed

¹<https://grouplens.org/datasets/movielens/>

²<https://jmcauley.ucsd.edu/data/amazon/>

Table 1: Experimental results of baseline models and the proposed framework on three datasets. Optimal value is displayed in bold, and the sub-optimal value is underlined.

Dataset	Metric	CMF	EMCDR	CoNet	DCDCSR	DTCDR	SSCDR	DisenCDR	DML	BiTGCF	GDCCDR	Fed-CLR	MFACDR
ML-1M	Recall@10	0.2540	0.2146	0.1498	0.2497	0.2373	0.0565	0.2176	0.2249	0.2208	<u>0.2611</u>	0.2337	0.2672
	MRR@10	0.4305	0.3656	0.1971	0.4164	0.3754	0.1701	0.3632	0.3428	0.3462	0.4565	0.4020	<u>0.4433</u>
	NDCG@10	0.2656	0.2269	0.1264	0.2536	0.2322	0.0803	0.1928	0.2173	0.2111	<u>0.2780</u>	0.2431	0.2785
	Hit@10	0.7253	0.6808	0.5451	0.7147	0.7031	0.3330	0.6370	0.6787	0.6850	0.7444	0.7041	<u>0.7423</u>
Book-Movie	Recall@10	0.0409	0.0351	0.0276	0.0387	0.0308	0.0256	0.0269	0.0249	0.0430	<u>0.0490</u>	0.0422	0.0540
	MRR@10	0.0250	0.0210	0.0177	0.0227	0.0174	0.0152	0.0171	0.0153	0.0271	<u>0.0313</u>	0.0270	0.0337
	NDCG@10	0.0239	0.0201	0.0120	0.0223	0.0172	0.0147	0.0163	0.0142	0.0257	<u>0.0289</u>	0.0276	0.0309
	Hit@10	0.0669	0.0611	0.0354	0.0613	0.0519	0.0427	0.0457	0.0439	0.0713	<u>0.0833</u>	0.0632	0.0888
Elect-Video	Recall@10	0.1085	0.0959	0.0535	0.0975	0.0507	0.0646	0.0877	0.0802	0.1193	<u>0.1216</u>	0.0906	0.1245
	MRR@10	0.0478	0.0395	0.0182	0.0447	0.0173	0.0240	0.0364	0.0287	0.0517	<u>0.0525</u>	0.0426	0.0573
	NDCG@10	0.0602	0.0508	0.0256	0.0552	0.0243	0.0324	0.0449	0.0312	0.0655	<u>0.0665</u>	0.0522	0.0691
	Hit@10	0.1165	0.1047	0.0581	0.1062	0.0550	0.0712	0.0937	0.0867	0.1296	<u>0.1326</u>	0.0988	0.1356

**Figure 2: Comparison of running time.**

MFACDR. Table 1 shows the experimental results. In addition, several top-performing models are selected for efficiency evaluation experiments. The results are shown in Figure 2.

From the observation of Table 1, the following conclusions can be drawn: First, the proposed MFACDR framework achieves superior performance across four evaluation metrics on all three datasets, demonstrating strong generalization and robustness in diverse cross-domain scenarios. Specifically, on the ML-1M dataset, MFACDR performs comparably to the sub-optimal model (GDCCDR). However, the running time comparison in Figure 2 shows that MFACDR is approximately 70% faster than GDCCDR, indicating a clear advantage in efficiency.

Second, compared with other baseline models, graph-based models (e.g., BiTGCF and GDCCDR) generally show better recommendation performance in various datasets, indicating that it is useful to construct graph structures to capture the interactive relationships between users and items. Moreover, GDCCDR, Fed-CLR, and MFACDR all incorporate contrastive learning as an auxiliary task. The competitive results of GDCCDR and Fed-CLR compared with non-contrastive baselines suggest that contrastive learning is effective in enhancing cross-domain representation alignment.

Finally, although DCDCSR and EMCDR do not utilize graph structures or contrastive learning, they still achieve above-average performance on ML-1M and Book-Movie. Both methods first learn intra-domain feature representations fully before mapping them, which highlights that sufficient intra-domain learning is critical for cross-domain recommendation.

4.3 Ablation Study (Q2)

To address the research question Q2, the ablation study is performed to demonstrate the effectiveness of different components in

Table 2: Experimental results of different variants of the proposed MFACDR. Optimal values are displayed in bold.

Dataset	Metric	w/o DADR	w/o meta	w/o rela	MFACDR
ML-1M	Recall@10	0.2580	0.2533	0.2480	0.2672
	MRR@10	0.4258	0.4289	0.4296	0.4433
	NDCG@10	0.2663	0.2639	0.2623	0.2785
	Hit@10	0.7211	0.7179	0.7094	0.7423
Book-Movie	Recall@10	0.0522	0.0528	0.0532	0.0540
	MRR@10	0.0332	0.0332	0.0334	0.0337
	NDCG@10	0.0311	0.0314	0.0320	0.0329
	Hit@10	0.0867	0.0876	0.0881	0.0888
Elect-Video	Recall@10	0.1134	0.1141	0.1197	0.1245
	MRR@10	0.0490	0.0503	0.0496	0.0573
	NDCG@10	0.0625	0.0630	0.0621	0.0691
	Hit@10	0.1240	0.1252	0.1292	0.1356

the proposed framework MFACDR in the cross-domain recommendation task. Three different variants are designed, as shown below. **MFACDR w/o DADR**: The DADR mechanism is removed from the proposed MFACDR. **MFACDR w/o meta**: The meta-learning process is removed from the proposed MFACDR. **MFACDR w/o rela**: The relation-based alignment is removed from the proposed MFACDR.

The experimental results of different variants of the MFACDR on three datasets are shown in Table 2. From the observation of Table 2, the following conclusions can be drawn: First, removing the DADR results in an overall degradation of all metrics on three datasets. This validates that dynamically combining historical and new data in a dual-stage replay manner can effectively alleviate catastrophic forgetting, enhance fine-grained representation learning. Similarly, removing the meta-learning process or relation-based alignment leads to a drop in all metrics. This confirms the effectiveness of meta-updates on non-overlapping data and relation-based alignment for cross-domain consistency. Overall, the proposed MFACDR consistently outperforms all ablated versions, indicating that these components jointly improve the framework.

4.4 Model Robustness Study (Q3)

To further investigate the robustness of the proposed framework (Q3), two types of experiments were conducted in comparison with two sub-optimal models, BiTGCF and GDCCDR. **Top-K Robustness**: This experiment was performed on the Elect-Video dataset

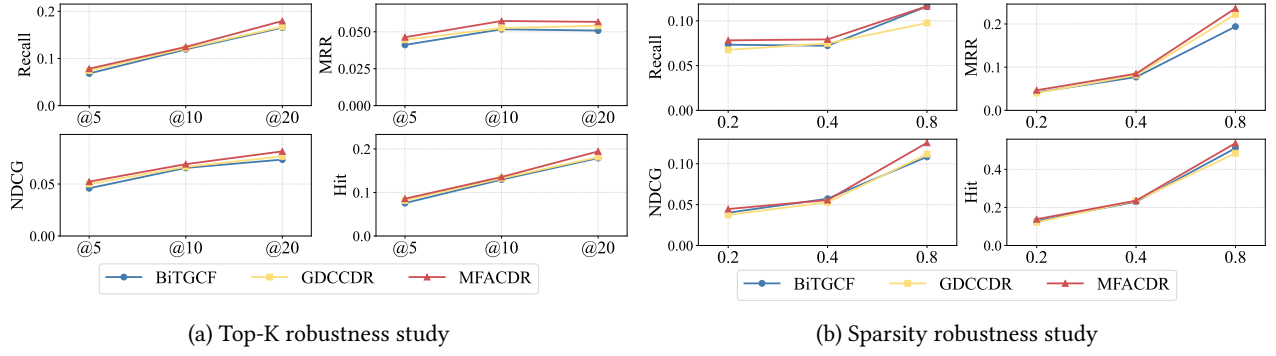


Figure 3: Results of robustness study

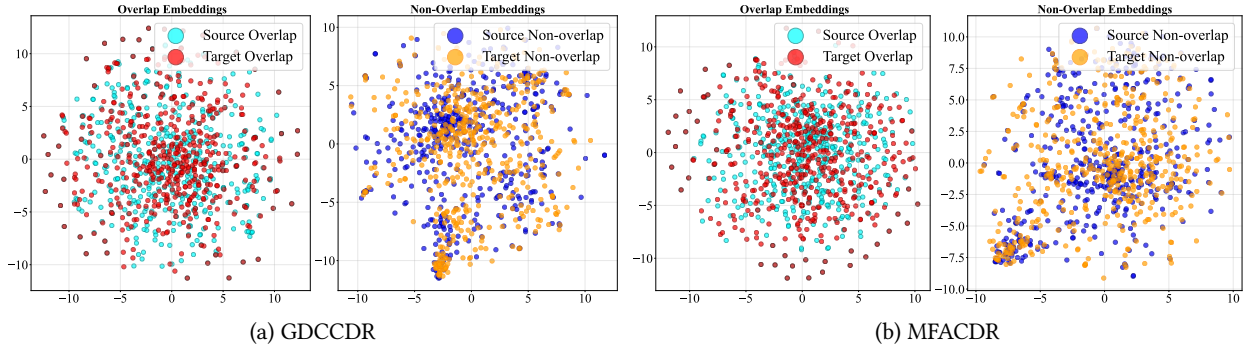


Figure 4: User embedding distribution using t-SNE on dataset Elect-Video

under three recommendation scenarios ($k = 5, 10$, and 20). As illustrated in Figure 3(a), MFACDR consistently outperformed the baselines across all Top-K settings. **Sparsity Robustness:** This experiment was conducted on the M1-ML dataset by varying the proportion of retained interaction data to 20%, 40%, and 80% of the original dataset. As illustrated in Figure 3(b), MFACDR maintained superior performance across all sparsity levels. This consistent outperformance confirms the robustness of the MFACDR framework. The framework’s ability to retain a performance advantage across varying recommendation list sizes indicates that its learned representations are not only highly effective but also stable.

4.5 Visualization of Alignment (Q4)

To address the research question Q4, the cross-domain alignment of users within the MFACDR and the suboptimal model GDCCDR was visualized on the Elect-Video dataset. Specifically, user embeddings were randomly sampled, including 500 overlapping users and 500 non-overlapping users. These high-dimensional embeddings were then projected into a two-dimensional space using the t-SNE, with the results shown in Figure 4. From Figure 4, we can see that when the alignment effects of overlapping users are similar, the alignment effect of non-overlapping users of MFACDR is better. This visualization demonstrates the effectiveness of the MFA method. This method allows the model to learn alignment knowledge from a small number of overlapping users and generalize it to a much larger population of non-overlapping users, thereby achieving effective cross-domain user representation alignment.

5 Conclusion and Future Work

In this study, a new framework named MFACDR is proposed to address the challenges of insufficient use of source domain data and coarse-grained representation learning in cross-domain recommendation. By integrating MFT method, MFACDR enables the model to autonomously discover flexible cross-domain alignment rules through inner–outer loop optimization, thereby enhancing adaptability to diverse cross-domain scenarios. Furthermore, the dual-stage adaptive data replay mechanism enables fine-grained cross-domain adaptability, alleviates catastrophic forgetting, and mitigates the risk of negative transfer. Experimental results on three real-world datasets have demonstrated the superior performance and robustness of MFACDR over state-of-the-art baselines. Ablation studies have demonstrated the effectiveness of the components proposed in this framework. These findings confirm the potential of combining meta-learning with adaptive data replay to enhance knowledge transfer in cross-domain scenarios and alleviate data sparsity. In future work, additional information sources such as textual descriptions and images could be incorporated to enrich domain knowledge. Moreover, large language models may be leveraged to process this multi-modal information, thereby enhancing semantic alignment between domains.

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A Theoretical Proof for Dual-Stage Adaptive Data Replay

A.1 Setup

Let θ_t denote the model parameters at training step t , and let $\mathcal{L}_t(\theta)$ be the loss function at that step. As shown in Eqs.(12) and (13), the loss consists of three components:

$$\mathcal{L}_t^X(\theta) = \mathcal{L}_{rec}^X(\theta) + \lambda_{EWC} \cdot \mathcal{L}_{EWC}^X(\theta) + \lambda_{GAN} \cdot \mathcal{L}_{GAN}^X(\theta), X \in \{S, T\}.$$

Since it is applicable to both domains, we henceforth abbreviate by $\mathcal{L}_t$, $\mathcal{L}_{rec}$, etc.

A.2 Assumptions

(A1) Lipschitz-smooth gradient. The gradient of $\mathcal{L}_{rec}$ is Lipschitz continuous with constant $L > 0$ [5]:

$$\|\nabla \mathcal{L}_{rec}(\theta) - \nabla \mathcal{L}_{rec}(\theta')\| \leq L \|\theta - \theta'\| \quad (\text{A.1})$$

where L controls the smoothness of gradient changes.

(A2) EWC Fisher boundedness. The Fisher information matrix F used for EWC is positive semi-definite, and each parameter's importance F_i is bounded above by $c > 0$ (i.e., c upper-bounds the maximum diagonal entry of F).

(A3) Replay moment alignment. The generator produces embeddings whose second-order moments are aligned with the empirical distribution of historical data, achieved via a bounded divergence regularizer (e.g., center-aware alignment). The replay generator generates a composite embedding $\tilde{e}$ such that for the corresponding real embedding e , we have $\|\tilde{e} - e\| \leq \delta$. Under these assumptions, one-step gradient descent updates are given by:

$$\theta_{t+1} = \theta_t - \eta \nabla \mathcal{L}_t(\theta_t) \quad (\text{A.2})$$

where η is the learning rate, θ_{t+1} are the new parameters after a gradient step, and $\nabla \mathcal{L}_t(\theta_t)$ is the gradient at the current parameters θ_t .

A.3 Lemma 1: Descent and Bounded Updates

According to the descent lemma (since the loss is Lipschitz smooth), for any θ, θ' :

$$\mathcal{L}_t(\theta') \leq \mathcal{L}_t(\theta) + \langle \nabla \mathcal{L}_t(\theta), \theta' - \theta \rangle + \frac{L}{2} \|\theta' - \theta\|^2 \quad (\text{A.3})$$

Substituting θ_t and θ_{t+1} from Eq. (A.2), we have $\langle \nabla \mathcal{L}_t(\theta_t), \theta_{t+1} - \theta_t \rangle = -\eta \|\nabla \mathcal{L}_t(\theta_t)\|^2$ and $\|\theta_{t+1} - \theta_t\|^2 = \eta^2 \|\nabla \mathcal{L}_t(\theta_t)\|^2$. Plugging the above into Eq. (A.3) yields:

$$\mathcal{L}_t(\theta_{t+1}) \leq \mathcal{L}_t(\theta_t) - \eta \|\nabla \mathcal{L}_t(\theta_t)\|^2 + \frac{L\eta^2}{2} \|\nabla \mathcal{L}_t(\theta_t)\|^2 \quad (\text{A.4})$$

or equivalently:

$$\mathcal{L}_t(\theta_{t+1}) \leq \mathcal{L}_t(\theta_t) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla \mathcal{L}_t(\theta_t)\|^2 \quad (\text{A.5})$$

Thus, the loss is guaranteed to decrease if the learning rate satisfies $0 < \eta < 2/L$. The training process maintains bounded parameter updates and can converge with the addition of EWC.

A.4 Lemma 2: Replay-induced Gradient Drift

In DADR, Consider the recommendation loss $\mathcal{L}_{rec}(\theta; e)$, whose gradient with respect to model parameters θ depends on the input embedding e . Define that:

$$g(\theta; e) \triangleq \nabla_{\theta} \mathcal{L}_{rec}(\theta; e) \quad (\text{A.6})$$

According to (A3), we know that the replay mechanism produces synthetic embeddings $\tilde{e}$ that satisfy a bounded replay error on the corresponding real embeddings e , i.e., $\|\tilde{e} - e\| \leq \delta$.

Furthermore, suppose that the gradient mapping $g(\theta; e)$ is Lipschitz continuous, such that for fixed θ there exists a constant $L > 0$ satisfying:

$$\|g(\theta; e_1) - g(\theta; e_2)\| \leq L \|e_1 - e_2\|, \quad \forall e_1, e_2 \quad (\text{A.7})$$

Then, the gradient deviation induced by replayed embeddings is bounded by:

$$\|g(\theta_t; \tilde{e}) - g(\theta_t; e)\| \leq L \|\tilde{e} - e\| \leq L\delta \quad (\text{A.8})$$

Therefore, the replay mechanism will not destabilize training as long as the generator maintains δ -closeness to the empirical embedding distribution.

A.5 Corollary: Combined Update Drift and Stability

As the EWC regularization term given in Eq.(11), its gradient satisfies:

$$\nabla_{\theta} \mathcal{L}_{EWC}(\theta) = F(\theta - \theta^*), \quad \|\nabla_{\theta} \mathcal{L}_{EWC}(\theta)\| \leq c \|\theta - \theta^*\| \quad (\text{A.9})$$

Combining the recommendation gradient, the drift caused by replay, and the EWC term, the parameter update magnitude can be limited as:

$$\|\theta_{t+1} - \theta_t\| = \eta \|\nabla \mathcal{L}_t(\theta_t)\| \leq \eta \left(\|\nabla \mathcal{L}_{rec}(e)\| + L\delta + \lambda_{EWC} c \|\theta_t - \theta^*\| \right) \quad (\text{A.10})$$

where the GAN-related gradient is assumed to be bounded under (A3) and absorbed into the constant terms.

This means that the optimized trajectory remains within the stable neighborhood of the previously learned solution. Therefore, the framework can gradually integrate new data; even when non-real samples are added, the overall update trajectory is controllable, and the addition of the DADR mechanism does not affect overall stability and convergence.

B Theoretical Proof for Meta-learning Driven Few-shot Knowledge Transfer

B.1 Setup

Let $\mathcal{E}_o$ denote overlapping (anchor) data and $\mathcal{E}_{non}$ denote non-overlapping data. Inner/outer parameters are denoted by p^{inner} and p^{outer} , respectively. Define the inner alignment loss $\mathcal{L}_{align}^{inner}$ and the outer alignment loss $\mathcal{L}_{align}^{outer}$. Set $F_{out}(p) := \mathcal{L}_{align}^{outer}(p; \mathcal{E}_{non})$, then:

$$g_{in}(p) := \nabla_{p^{outer}} \mathcal{L}_{align}^{inner}(p; \mathcal{E}_o) \quad (\text{B.1})$$

$$g_{out}(p) := \nabla_{p^{outer}} F_{out}(p) \quad (\text{B.2})$$

Followed by Eq.(20) and Eq.(27):

Inner update:

$$\tilde{p}^{inner} = p^{inner} - \alpha \nabla_{p^{inner}} \mathcal{L}_{align}^{inner}(p^{inner}; \mathcal{E}_o)$$

Outer objective:

$$\begin{aligned} \min_{p^{outer}} \mathcal{L}_{align}^{outer}(\tilde{p}^{outer}; \mathcal{E}_{non}) = \\ \min_{p^{outer}} \mathcal{L}_{align}^{outer}(p^{outer} - \eta \nabla_{p^{outer}} \mathcal{L}_{align}^{inner}(p^{outer}; \mathcal{E}_o); \mathcal{E}_{non}) \end{aligned}$$

Specifically, the inner update adjusts the parameters by using the gradient step size of the inner alignment loss through anchor data. The outer objective then evaluates how this inner-driven update performs on non-overlapping users.

B.2 Assumptions

(A1) **Smoothness.** F_{out} is L_{out} -smooth.

(A2) **Gradient alignment.** There exists a constant $\kappa \in (0, 1]$ such that:

$$\langle g_{out}(p), g_{in}(p) \rangle \geq \kappa \|g_{out}(p)\| \|g_{in}(p)\| \quad (\text{B.3})$$

This assumption suggests that the inner gradient learned from overlapping (anchor) data is positively aligned with the outer gradient evaluated on non-overlapping users.

B.3 Lemma: One-step Descent

Consider the inner-driven update: $p^+ = p - \alpha g_{in}(p)$, where p^+ is the new parameter after one inner-driven update, and α is the learning rate.

Since F_{out} is L_{out} -smooth, the descent lemma yields:

$$\begin{aligned} F_{out}(p^+) &\leq F_{out}(p) + \langle \nabla F_{out}(p), p^+ - p \rangle + \frac{L_{out}}{2} \|p^+ - p\|^2 \\ &= F_{out}(p) - \alpha \langle g_{out}(p), g_{in}(p) \rangle + \frac{L_{out}}{2} \alpha^2 \|g_{in}(p)\|^2 \end{aligned} \quad (\text{B.4})$$

Applying the gradient alignment condition in Eq.(B.3), we can obtain that:

$$F_{out}(p^+) \leq F_{out}(p) - \alpha \kappa \|g_{out}(p)\| \|g_{in}(p)\| + \frac{L_{out}}{2} \alpha^2 \|g_{in}(p)\|^2 \quad (\text{B.5})$$

Therefore, any learning rate satisfying:

$$0 < \alpha < \frac{2\kappa \|g_{out}(p)\|}{L_{out} \|g_{in}(p)\|} \quad (\text{B.6})$$

guarantees $F_{out}(p^+) < F_{out}(p)$. A balanced choice that maximizes the guaranteed decrease is:

$$\alpha^* = \frac{\kappa \|g_{out}(p)\|}{L_{out} \|g_{in}(p)\|} \quad (\text{B.7})$$

Condition (B.3) ensures that the inner update direction learned from overlapping users does not conflict with the outer goals of non-overlapping users.

B.4 Corollary: Non-overlapping Users Become Aligned

Since $F_{out}(p) = \mathcal{L}_{align}^{outer}(p; \mathcal{E}_{non})$ is evaluated on non-overlapping pairs, the reduction of $F_{out}(p)$ directly means the alignment on non-overlapping data has been optimized. If there exists an alignment

discrepancy measure $L_{align}(p)$ that is upper-bounded by the outer alignment loss $F_{out}(p)$ up to a constant factor:

$$L_{align}(p) \leq c F_{out}(p)$$

for some constant $c > 0$, then the decrease of F_{out} ensured by Eqs.(B.5)–(B.6) implies that the alignment on non-overlapping users is improved:

$$L_{align}(p^+) < L_{align}(p)$$

Hence, the inner update driven by few-shot overlapping data promotes alignment on non-overlapping users, providing a theoretical justification for meta-learning driven few-shot knowledge transfer.

C Dataset detail

Table 3: Information of three CDR datasets.

Dataset		User	Item	Interaction	Sparsity
ML-1M	Source	6039	3308	835789	95.82%
	Target	944	1197	81705	92.77%
Book-Movie	Source	172730	160274	4689366	99.98%
	Target	33260	22243	843362	99.88%
Elect-Video	Source	150524	52030	1312570	99.98%
	Target	18814	8692	177572	99.89%

D Baselines

We compare the MFACDR with the following state-of-the-art models:

CMF[30]: A joint matrix factorization method that shares user embeddings between domains.

EMCDR[24]: Learns user embeddings via matrix factorization (MF) and transfers them from a source to a target domain with a neural network.

CoNet [11]: Simultaneously learns knowledge across two domains by transferring user-item interaction features.

DCDCSR[45]: Learns embeddings with matrix factorization and uses a deep neural networks for cross-domain mapping.

DTCDR [44]: Uses a dual-tower structure to learn both independent and shared features across two domains simultaneously.

SSCDR[13]: Trains a bridge function using a semi-supervised approach.

DisenCDR[3]: Disentangles domain-shared and domain-specific user representations via mutual-information regularizers to enable effective cross-domain knowledge transfer.

DML[17]: Utilizes dual learning with latent orthogonal mapping and metric learning to iteratively transfer knowledge between domains.

BITGCF [21]: Achieves bidirectional knowledge transfer between domains by using shared users as a bridge.

GDCCDR[20]: Combines graph neural networks and contrastive learning to enhance the consistency of cross-domain representations.

Fed-CLR[36]: Combines federated learning with contrastive learning and utilizes a dual constraint mechanism to accelerate convergence.